

Chemistry of Shampoo

Shampoo is one of the hair care product. These are hair cleanser. Shampoo are used for the removal of oils, dirt, dandruff, environmental pollutants and other contaminant particles that gradually build up in hair.

Functions of Shampoo

1: Cleaning agents

- Shampoos should be mild, effective, without causing irritation.
- Should be used daily or on alternate days as needed
- They remove dust and excess oil from the hair

2: Anti seborrhoeic agents

- Preventing excessive secretion of sebum
- They have cytostatic effect on cells of the epidermis and follicular epithelium thus reducing corneocyte production

3: Anti dandruff agents

- These treat dandruff due to fungi like
- Rapidly relieves scaling and pruritis which are associated with fungal infections.

4: Keratolytic agents

- They remove ointment, pastes which are used in the treatment of psoriasis.
- They also remove hard scales from the scalp

Qualities of Shampoo

- 1: Easy spreading
- 2: Easy rinsing
- 3: Should completely remove dust or soil, and excessive sebum
- 4: No damage to hair.
- 5: Pleasant fragrance
- 6: Low toxicity
- 7: Good biodegradability
- 8: Good finish after washing hair
- 9: Minimal skin/eye irritation

- 10: Feels thick and Creamy.
- 11: Good foaming ability.
- 12: Slightly acidic (pH-7)

Composition

Following are the main components of Shampoo.

- 1: Surfactant (Cleaning agents)
- 2: Preservatives
- 3: Germicidal and antiodandruff agents
- 4: Conditioning agents.
- 5: Thickening agent.
- 6: Colour additives.
- 7: Perfumes or fragrance.
- 8: pH adjuster.
- 9: Pearl Scent agent.
- 10: Clarifying agents

Surfactants (cleaning agents)

Surfactants are prime ingredients of all shampoos and are responsible for cleaning action, lathering properties

The major types of surfactants are.

- i - Anionic (ii) Cationic (iii) Non ionic (iv) Amphoteric

1: Anionic Surfactants

- Anionic surfactants carry a -ve charge when ionized
- It provides lots of lather, excellent cleanings and foaming
- The most commonly used anionic are
Sod. lauryl sulphate
- Their disadvantage is that they can be harsh and irritating to the scalp

(ii) Cationic Surfactants

- Cationic surfactants carry +ve charge when ionized.
- They are used less frequently due to their dangerous threat to eyes if used in large quantities.
- They are not easily removed during the rinsing.
- Polyquarternium-10 is one of the most common cationic conditioners.

(iii) Non Ionic Surfactants

- Non ionic surfactant has no charge to the molecule.
- Not used as a cleansing agent, but are often used in combination with the primary cleanser.
- They can strip the hair and lead to scalp irritation due to excessive defatting.
- They include laureth-3 or 4, cocamide diethanolamine.

(iv) Amphoteric Surfactants.

- It carries both +ve and -ve charges when ionized.
- They are very useful for decreasing the irritation and good quality lather is produced.
- Mostly amphoteric shampoo surfactants are used in baby shampoos, because they are gentle and won't burn the eyes.

Examples: Cocamido propyl betaine.
cocamide betaine

Functions of surfactants in Shampoo

- i) Cleaning / cleansing.
- Its Primary function is cleansing. Surfactant system must work quickly at low temperature and must be effective in hard and soft water, able to remove lipids and other soils and must not leave any residues.
 - It must be non-toxic and reasonably non-irritant to skin and eyes.

2) Foaming

Foam formation is also a function of the primary surfactants. Some additional materials may depress the foam and make it creamier.

3) Conditioning

They help in improving hair condition and leave the hair easy to manage and silky.

4) Viscosity:

Products must have sufficient viscosity to stay on the palm of hand before application but must not come out of bottles as a globular lump.

Sod. chloride is the material of common choice.

Preservatives

Bacterial attack may lead to break detergency in shampoos results in discoloration of product.

Sod. benzoate is ~~another~~ a preservative used in shampoos.

It kills bacteria, fungi & yeast,

Dimethyl Iodimethyl hydantion, 2-bromo-2-nitropropane-1,3 diol (bactericide)

Germicidal & antidandruff agents.

These are used to prevent microbial infection and treat dandruff.

Germicide — quaternary ammonium compounds
Benzalkonium chloride

Anti dandruff. — Selenium sulphide
Cadmium sulphide etc

Conditioning agents

Give special conditioning effect to hair.

Example: lanolin oils, herbal extract, Egg, Amino acids, Lecithin

Thickening agent

These are used to make shampoo viscous for easy handling and minimize wastage while pouring.

Examples: Alginates, PVA, colloidal Silicates
sulphonated castor oil, sulphonated olive oil

Color Additives

- These are used by some manufacturer only and only certified colors are used
- These are used to produce attractive impact
- Colors should be water soluble

Perfumes

- These are used to have an after-use fragrance on the hair
- Various perfumes are used in concentration of 0.3-1.0%.

pH balance

- Most shampoos are made slightly acidic to keep the cuticle smooth and lying flat on the hair root.
- Citric acid is added to acidify the shampoo.
- As the shampoo mixes with water or mixes with dirt on the hair it can become less acidic as acids mix with alkaline water or dirt.
- A typical buffering agent used in shampoo is sodium citrate

Pearl Scent agent

These are used to shine the hairs and brighten the hairs

Examples 4-Methyl coumarins: used in conc. of 0.2-1%.

and generally used in pH 4.5-6

Clarifying agents

These are used to clear the shampoo, and used to improve transparency ↑

Example: Ethanol, Isopropyl alcohol, Phosphates etc

Types of Shampoo

1. Powder Shampoo : $\text{NaHCO}_3 = 50\%$, disodium phosphate = 20%
Soap powder = 30% . Perfumes .

2. Clear Liquid Shampoo Triethanolamine lauryl Sulphate = 45%
Coconut monoethanolamine = 2%
water = 53% , Perfume, color, preservative

3. oil Shampoo Sulphonated olive oil = 16%
Sulphonated Castor oil = 16%
Water = 68%

Perfumes, color, preservative

4. Antidandruff Shampoo Thymol = 0.05%
Methanol = 0.10%
Camphor = 0.10%

Triethanolamine lauryl sulphate = 50%
Water = 49.75%
Perfume

Herbs used in Shampoo

Aloe, Marigold, Arnica, Rosemary, Chamomile, Licorice are used in various shampoos

Evaluation of Shampoo

To evaluate the property/quality of shampoo followings are determine .

1) Determination of pH .

2) Determination of foam formation .

3) Foam quality and retention

4) Determination of relative viscosity

5) Dirt dispersion

6) Effect on skin and eyes .

7) Effect on hair